



Delivering Machine Learning Engineering in Scientific Research

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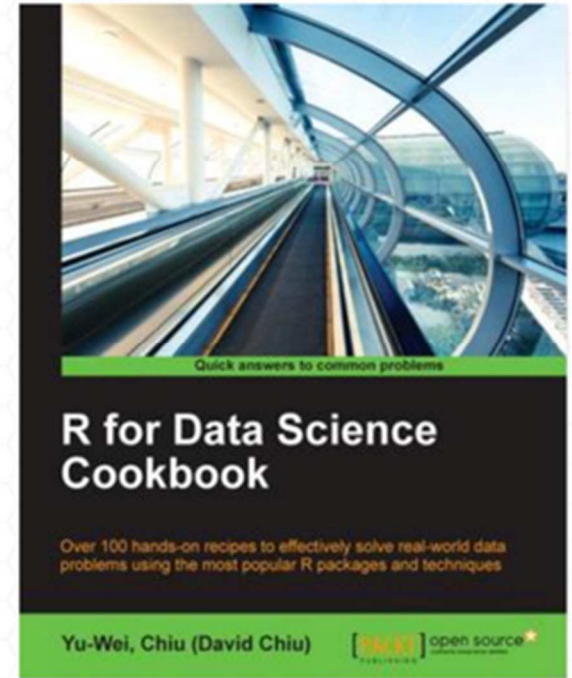
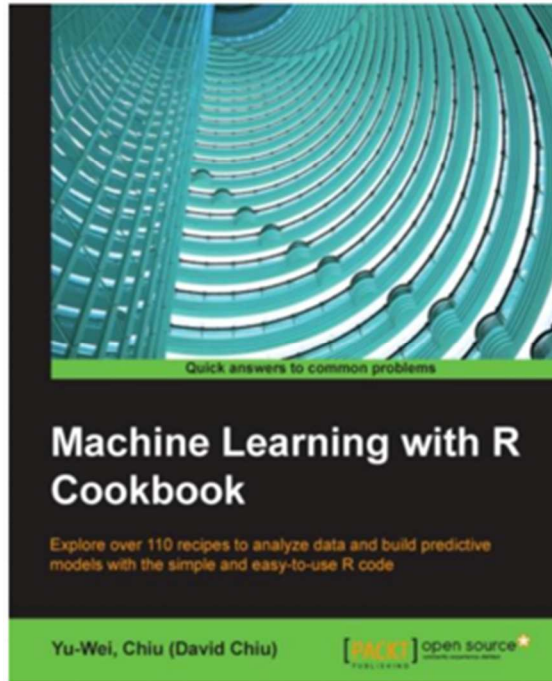
Phone: +886929094381

About Me



- Founder of LargitData Inc.
<http://www.largitdata.com/>
- Fan Page
<https://www.facebook.com/largitdata>
- R for Data Science Cookbook
<https://www.packtpub.com/big-data-and-business-intelligence/r-data-science-cookbook>
- Machine Learning With R Cookbook
<https://www.packtpub.com/big-data-and-business-intelligence/machine-learning-r-cookbook>

Machine Learning With R Cookbook & R for Data Science Cookbook



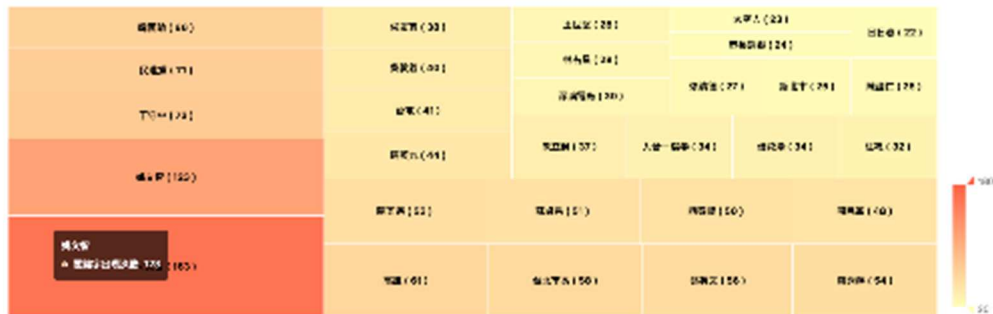
Author: Yu-Wei (David) Chiu



When Machine Learning
become a must in your
work or research

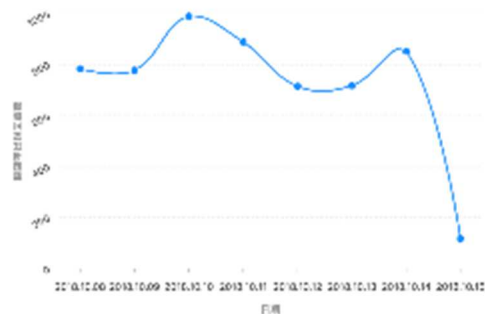
InfoMiner – A Realtime Opinion Analysis System

Find topics of the day



關鍵字搜尋結果分析

數量分析 來源分析 文章列表



分類分析

分類分析結果



回國學論 柯文哲批評天吃地罵罵沒事

柯文哲 丁守中 台北市長 陳文雄 新文化運動 李星峰 六合一選舉 連勝文 市選選舉 日日新

藍綠陣營 柯P反政：每天吃地罵沒事

丁守中：柯文哲不願接受民意調查 是市長選舉的隱憂

迴響文高票可中：不該舉辦 或像明紅綠化投發自標

日日本文藝界抗議 柯文哲冷靜不乾一稿

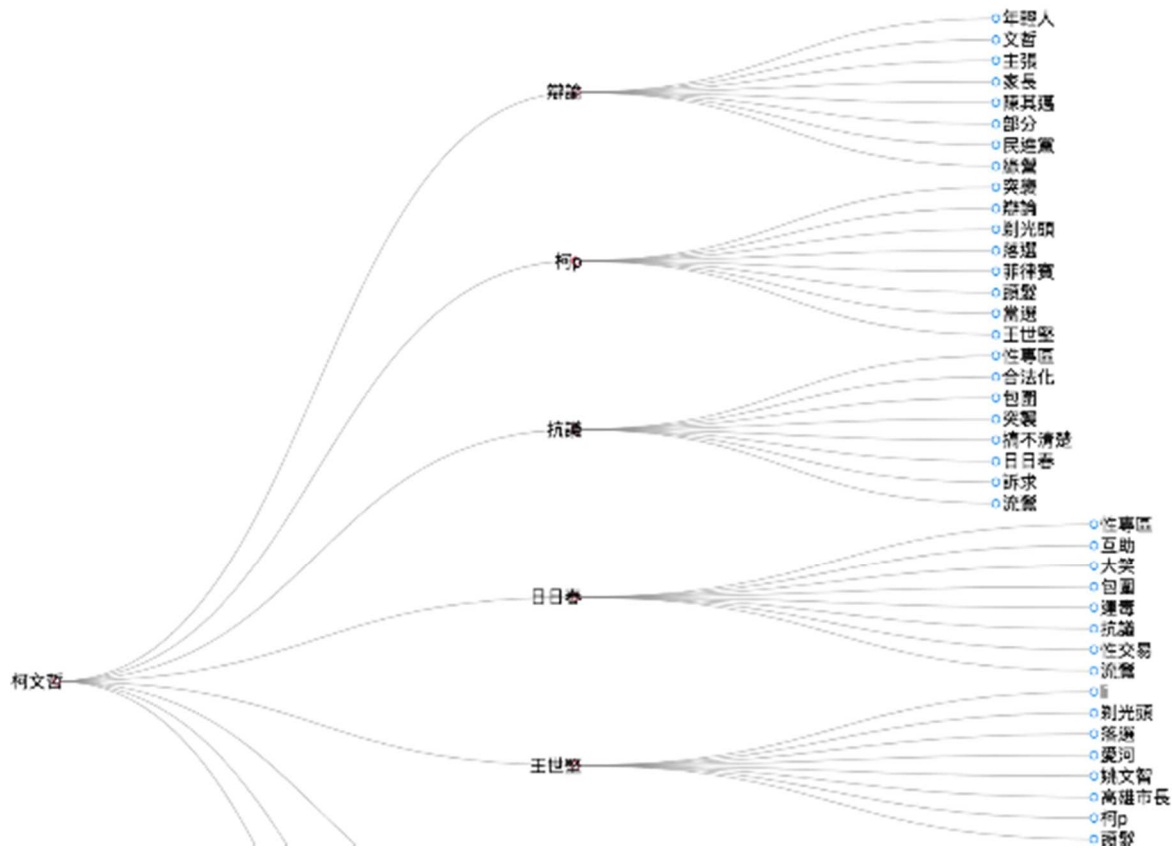
北市民選戰 柯區參戰後 選民價值相同

文章數: 123 | 贊賞分數: 137.3 | 相關文章

新聞

InfoMiner – A Realtime Opinion Analysis System

Automatically Discover Associated Topics



InfoMiner – A Realtime Opinion Analysis System

Text Summarization

排名	3	議題	盧秀燕競選總部成立 馬吳等天王到場		
文章數	123	聲量分數	131.7	關鍵字	盧秀燕, 馬英九, 國民黨, 吳敦義, 林佳龍, 台中市長, 民進黨, 九合一選舉, 王金平, 朱立倫
議題摘要	國民黨台中市市長參選人盧秀燕上午成立競選總部，前總統馬英九和國民黨主席吳敦義都到場支持，馬英九強調，台中市在這場選舉很重要。				
相關文章	新頭殼 newtalk: 競總成立馬吳朱王同框 盧秀燕：與東廠的執政黨直球對決 自由時報電子報: 全國走透透賣力站台 馬英九：任內丟政權、用輔選贖罪 6連霸市議員何敏誠拚連任 陳雷唱「猴賽雷」助拳 分析台中市長選戰 學者提林佳龍、盧秀燕勝負關鍵在... 行政院停建深澳電廠 馬英九批為救新北、台中選情 國民黨六都合體萬人造勢 高呼2020重返執政 抬轎也坐轎！鄭文燦跑競選總部成立大會 無黨議員也挺 盧秀燕競總成立 宣示與執政黨「直球對決」				

Data Analysis Language



R Language

Make data analysis pretty easy

- Data Munging
- Data Analysis
- Report Making

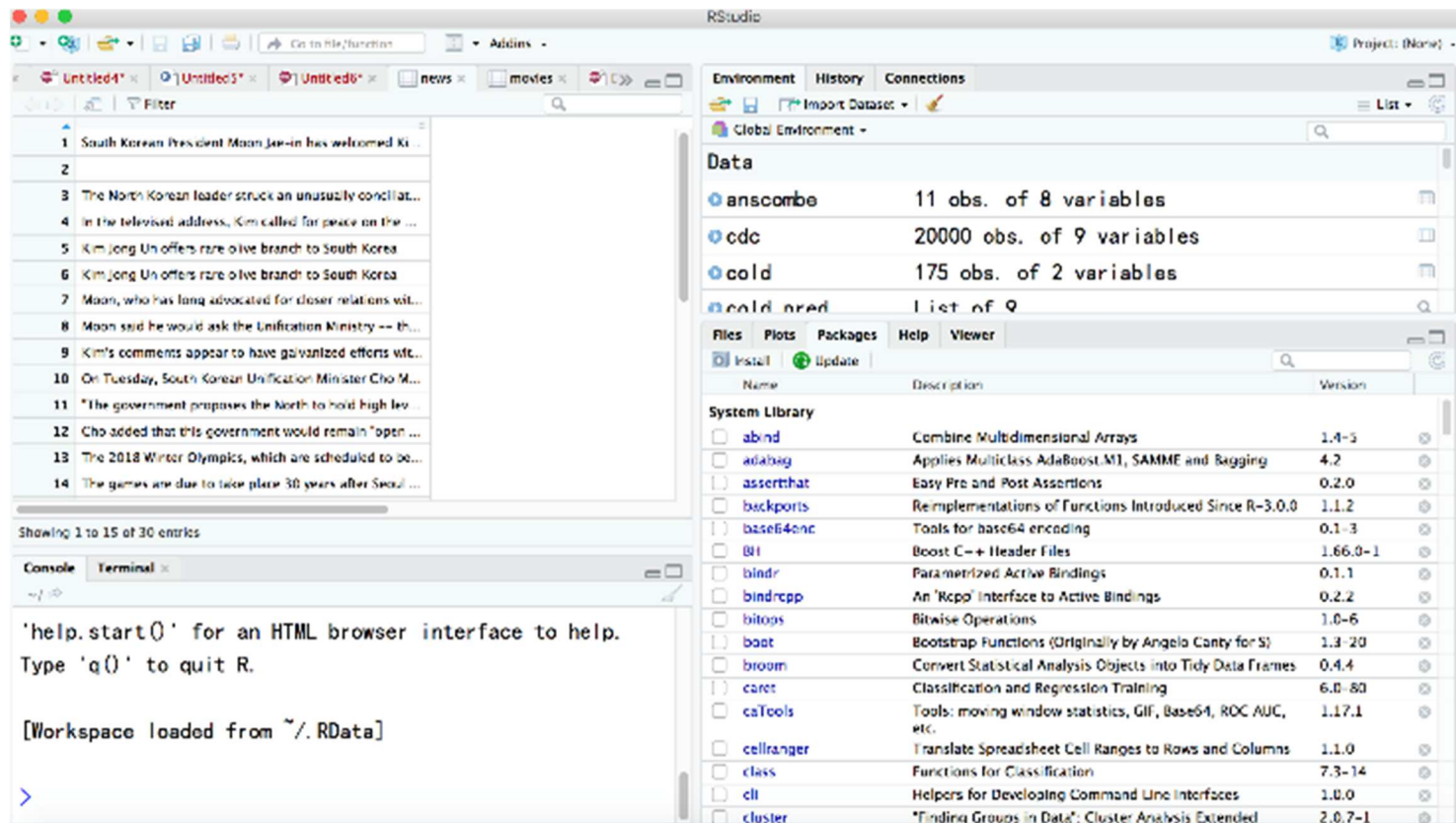
Many built-in and third-party modules

Open source and free

Easy to customize and expand



Develop a R program with RStudio



The screenshot displays the RStudio environment with the following components:

- Script Editor:** Contains a list of 14 news entries, such as "South Korean President Moon Jae-in has welcomed Ki..." and "The North Korean leader struck an unusually conciliat...".
- Environment Pane:** Shows the current data environment with three objects: `anscombe` (11 obs. of 8 variables), `cdc` (20000 obs. of 9 variables), and `cold` (175 obs. of 2 variables). Below this is a list of installed packages.
- Console:** Displays the output of the `help.start()` function, which opens an HTML browser interface to help. The message reads: `'help.start()' for an HTML browser interface to help. Type 'q()' to quit R.` Below this, it says `[Workspace loaded from ~/.RData]`.
- Package List:** A table of installed and available packages.

Name	Description	Version
☐ abind	Combine Multidimensional Arrays	1.4-5
☐ adabag	Applies Multiclass AdaBoost.M1, SAMME and Bagging	4.2
☐ assertthat	Easy Pre and Post Assertions	0.2.0
☐ backports	Reimplementations of Functions Introduced Since R-3.0.0	1.1.2
☐ base64enc	Tools for base64 encoding	0.1-3
☐ bit	Boost C++ Header Files	1.66.0-1
☐ bindr	Parameterized Active Bindings	0.1.1
☐ bindrapp	An 'Rcpp' interface to Active Bindings	0.2.2
☐ bitops	Bitwise Operations	1.0-6
☐ boot	Bootstrap Functions (Originally by Angelo Canty for S)	1.3-20
☐ broom	Convert Statistical Analysis Objects into Tidy Data Frames	0.4.4
☐ caret	Classification and Regression Training	6.0-80
☐ caTools	Tools: moving window statistics, GIF, Base64, ROC AUC, etc.	1.17.1
☐ cellranger	Translate Spreadsheet Cell Ranges to Rows and Columns	1.1.0
☐ class	Functions for Classification	7.3-14
☐ cli	Helpers for Developing Command Line Interfaces	1.0.0
☐ cluster	"Finding Groups in Data": Cluster Analysis Extended	2.0.7-1

Python Language

Cross platform

Many built-in and third-party modules

Easier to read and understand

Deep learning



Develop a python program with Jupyter Notebook

 jupyter Credit Card Fraud Detection Last Checkpoint: 07/22/2018 (autosaved)  Logout

File Edit View Insert Cell Kernel Help

Trusted

Python 3

        Run   Markdown 

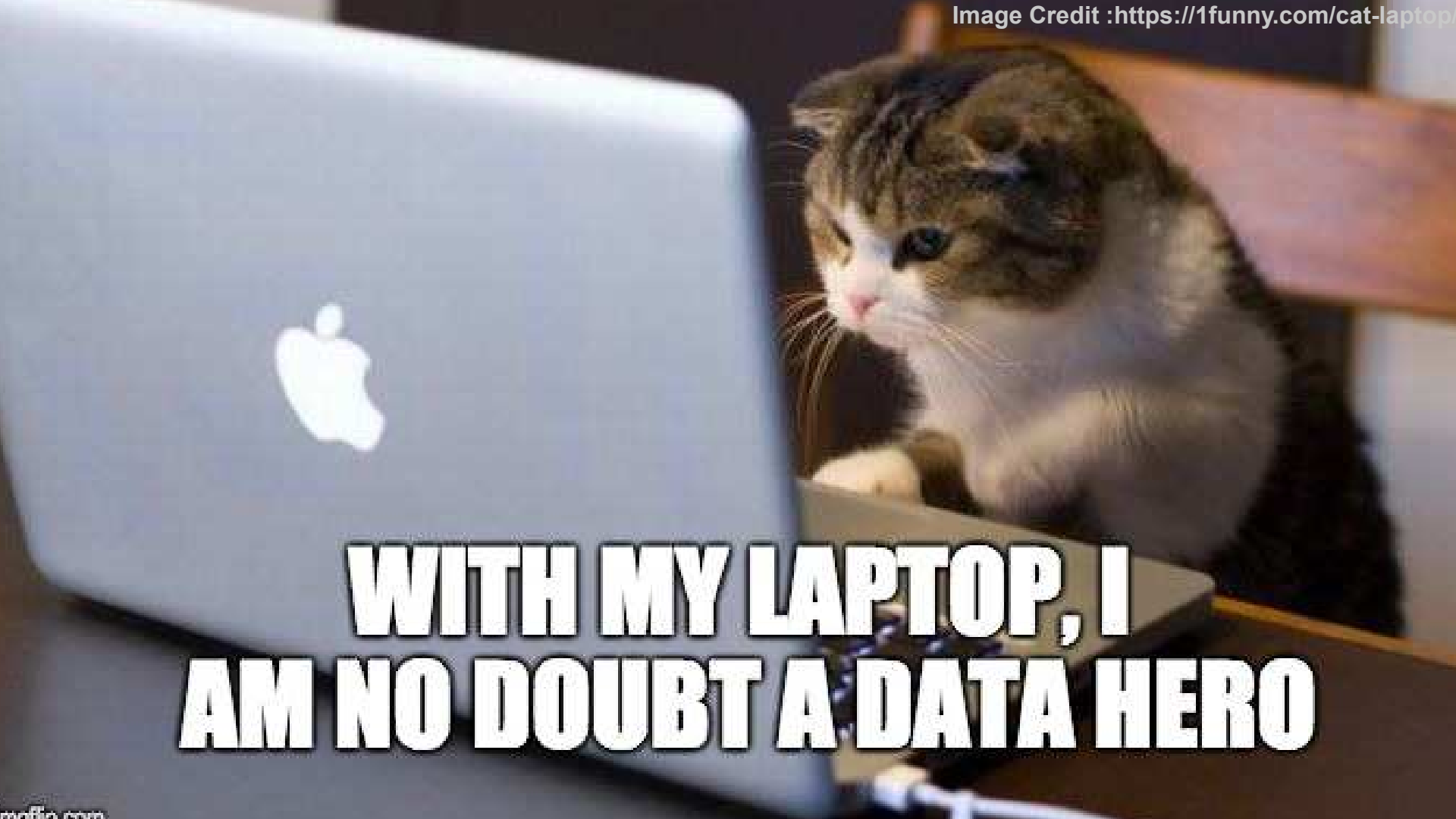
```
In [16]: nb_epoch = 10
batch_size = 32

autoencoder.compile(optimizer='adam',
                    loss='mean_squared_error',
                    metrics=['accuracy'])

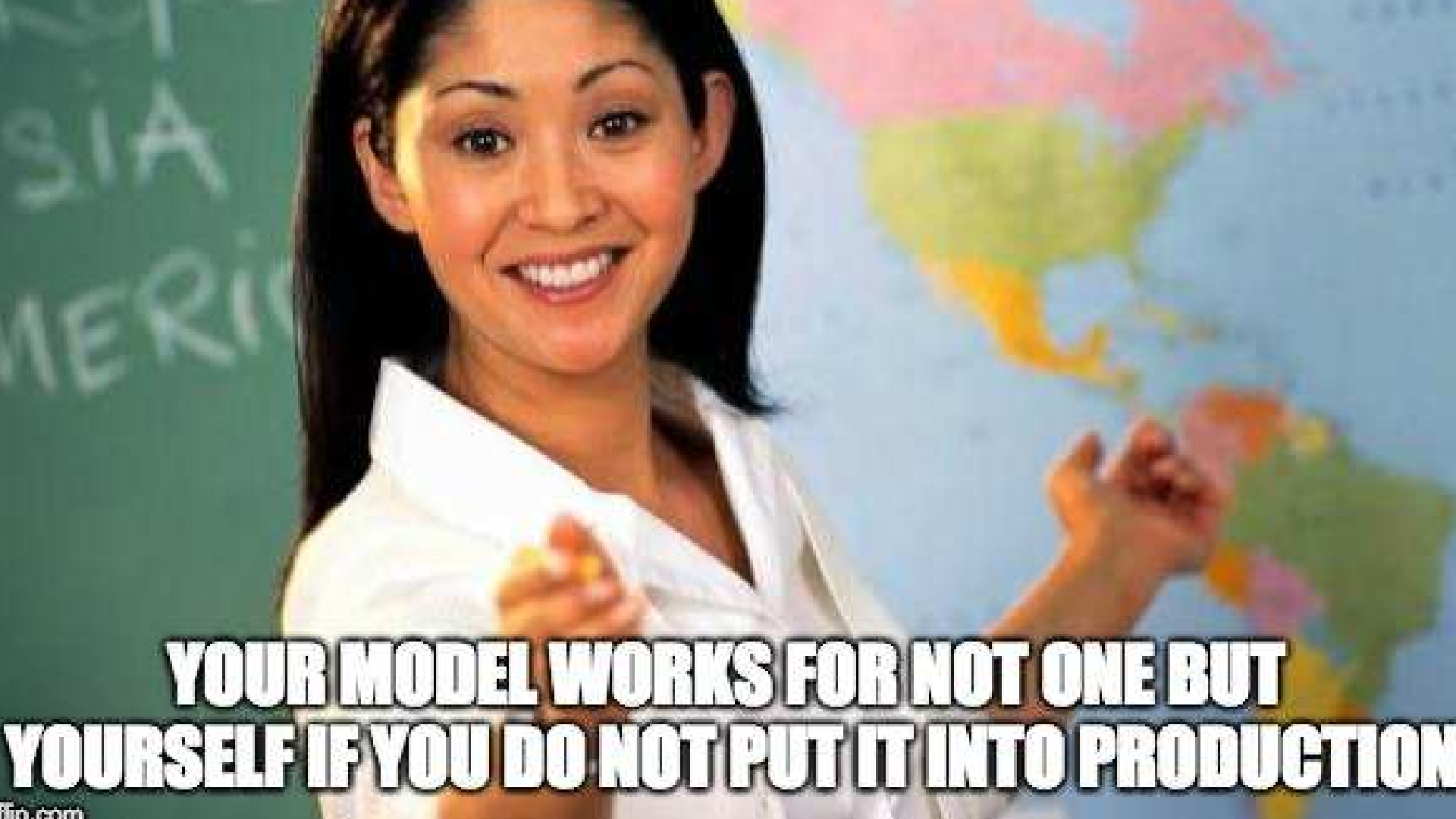
checkpointer = ModelCheckpoint(filepath="model.h5",
                              verbose=0,
                              save_best_only=True)

history = autoencoder.fit(X_train, X_train,
                        epochs=nb_epoch,
                        batch_size=batch_size,
                        shuffle=True,
                        validation_data=(X_test, X_test),
                        verbose=1,
                        callbacks=[checkpointer]).history
```

WARNING:tensorflow:From /Users/davidchiu/.pyenv/versions/3.6.2/lib/python3.6/site-packages/keras/backend/tensorflow_backend.py:1349: calling reduce_mean (from tensorflow.python.ops.math_ops) with keep_dims is deprecated and will be removed in a future version.
Instructions for updating:
keep_dims is deprecated, use keepdims instead

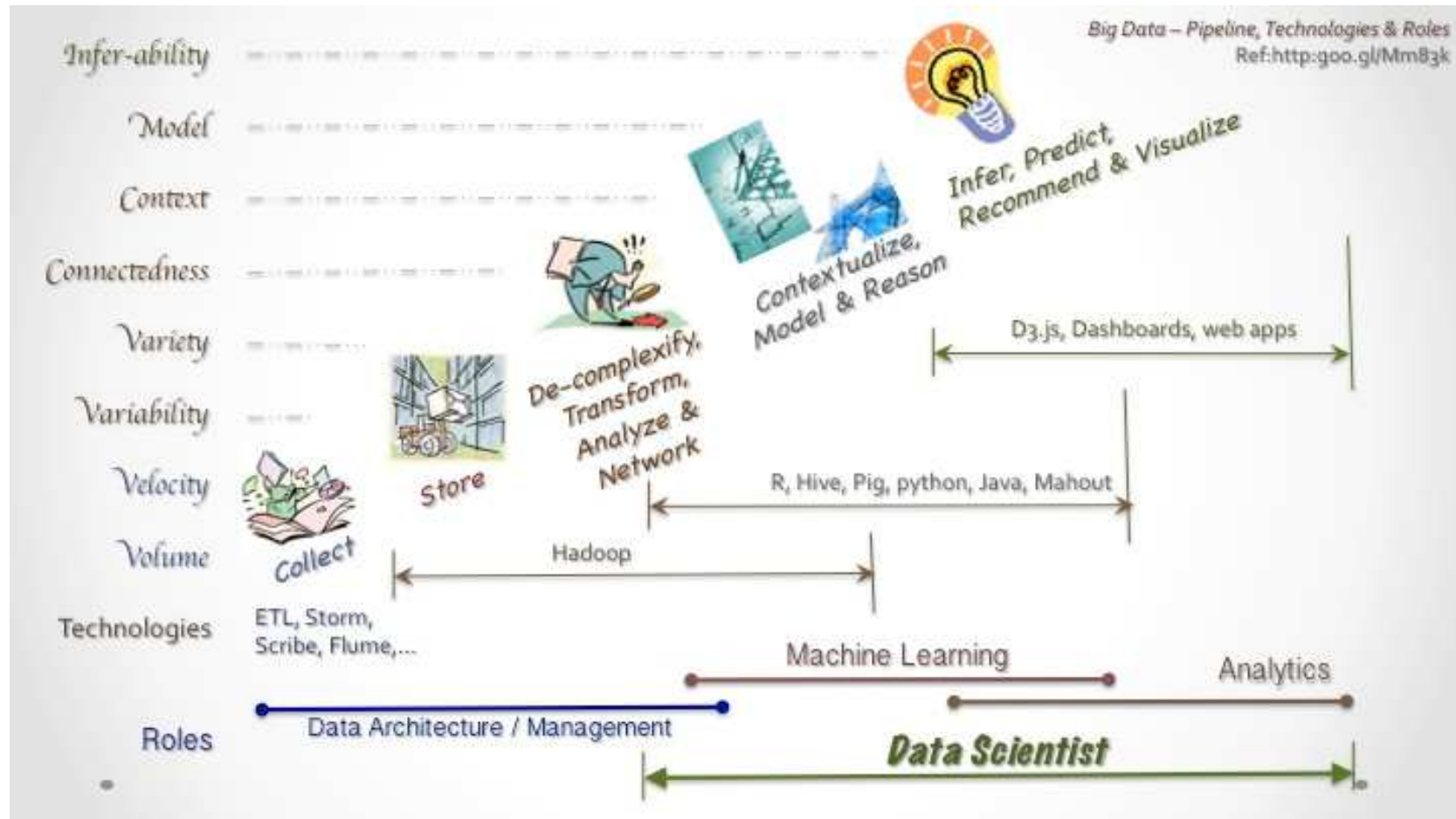


**WITH MY LAPTOP, I
AM NO DOUBT A DATA HERO**

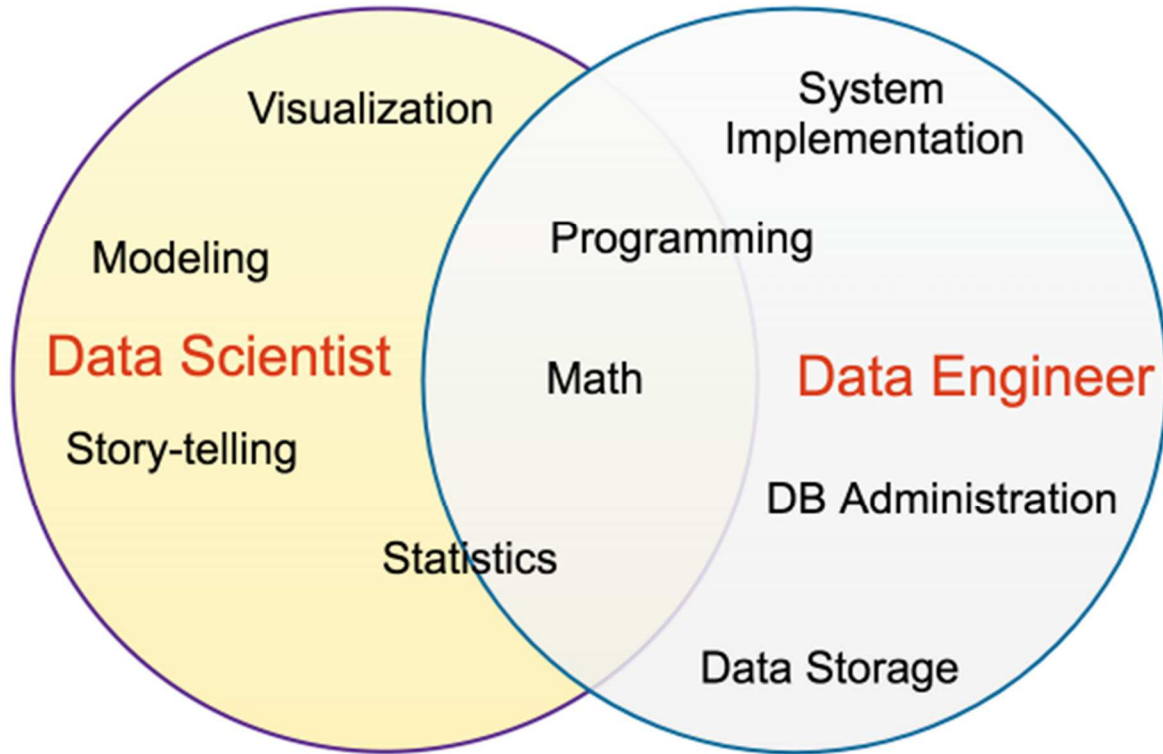


**YOUR MODEL WORKS FOR NOT ONE BUT
YOURSELF IF YOU DO NOT PUT IT INTO PRODUCTION**

Data science workflow



When Data Scientist Meets Data Engineer



Machine Learning Workflow

Data Engineering

- Acquisition
- Preprocessing
- Data Munging

Data Science

- EDA
- Model Training
- Model Testing

Production

- Reports & Dashboard
- Model Scoring
- API
- Real-time Prediction
- Deploying & Monitoring

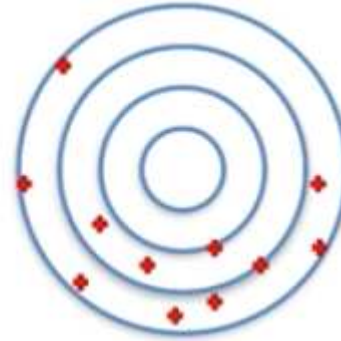


Model Testing

Bias-Variance Tradeoff



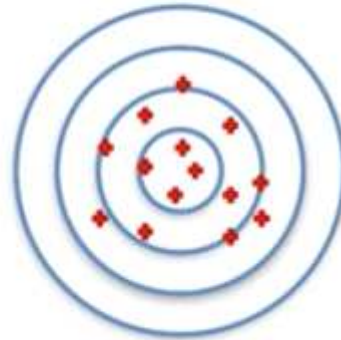
High Bias Low Variance



High Bias High Variance

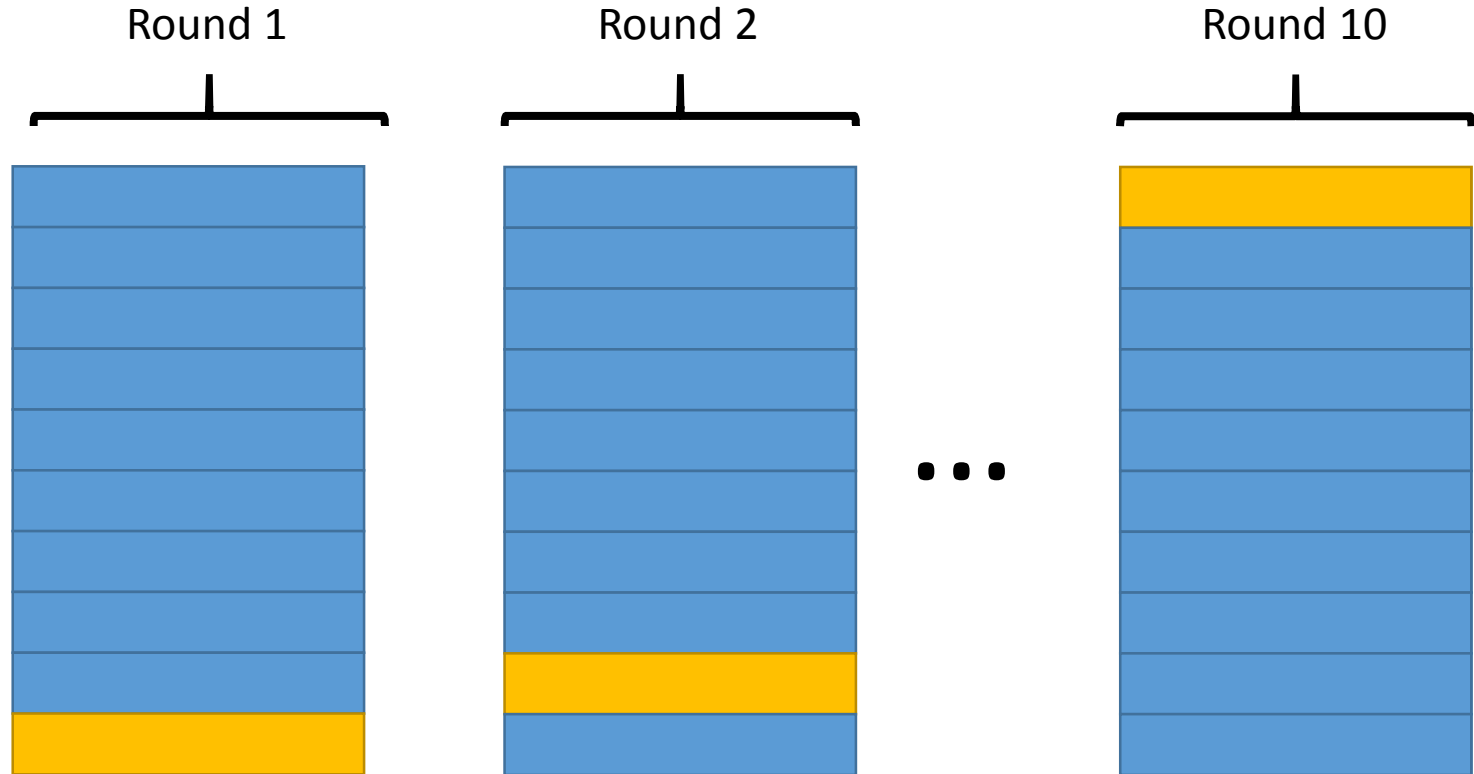


Low Bias Low Variance

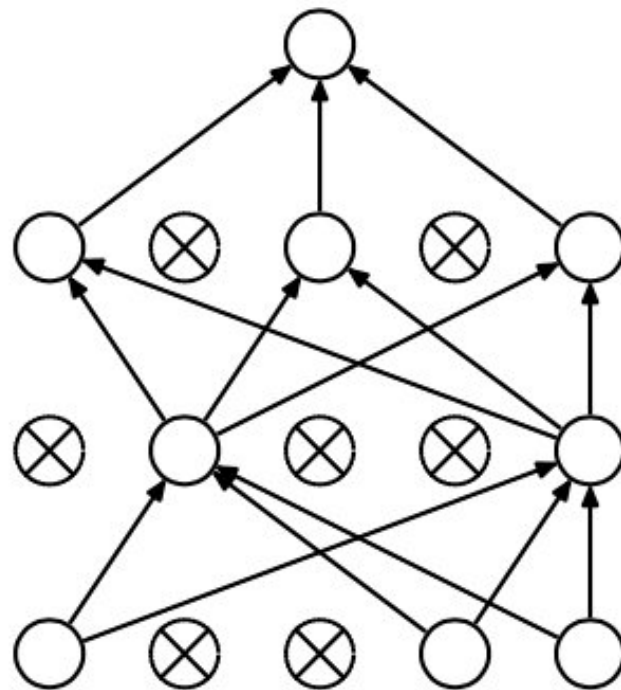
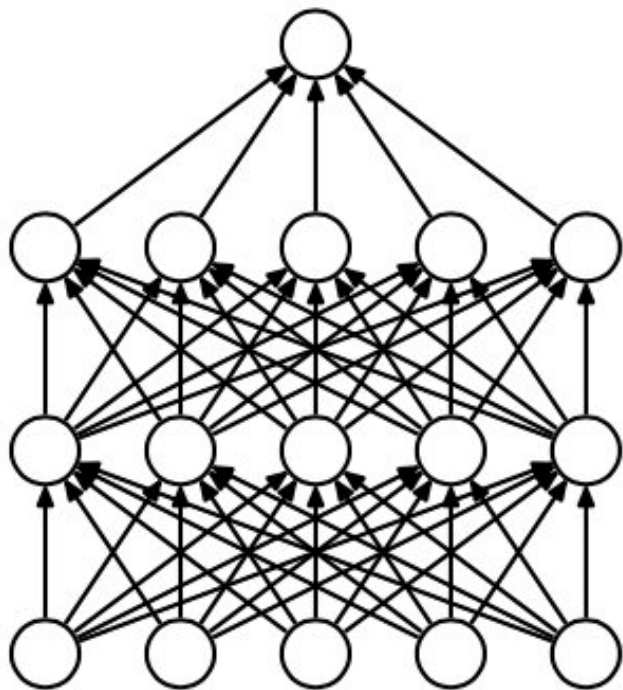


Low Bias High Variance

Cross Validation

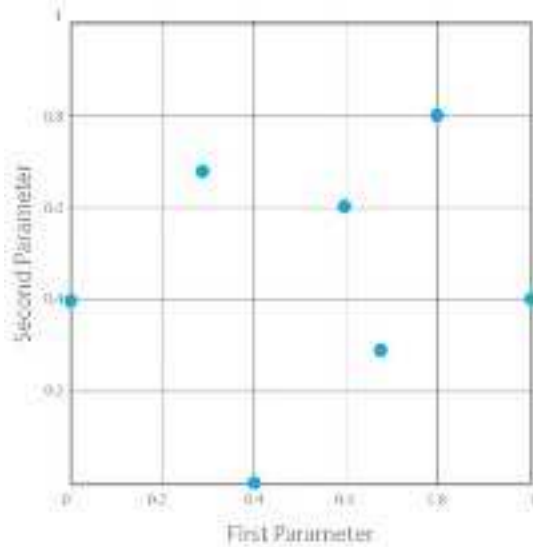


Prevent Overfitting

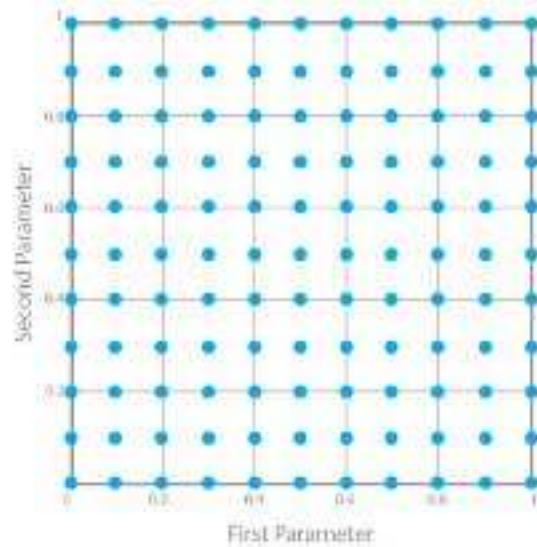


Model Tuning

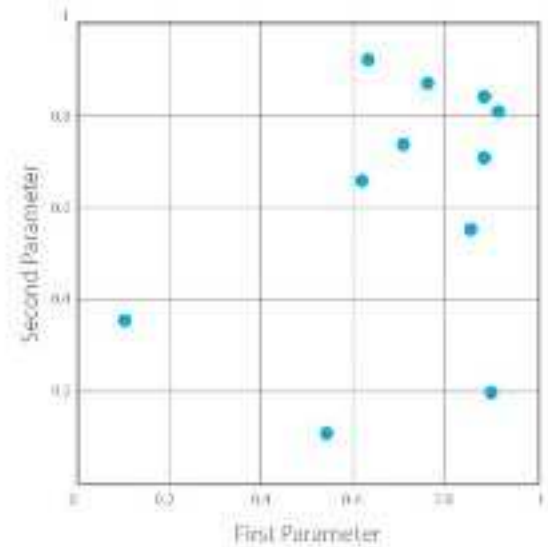
Manual Search



Grid Search



Random Search





Model Management

How to Persist Model?

Storing complete Python/R objects into files and later loading them back



Pickle



How to Exchange Model?

What if we need to load files from model built with different language



Pickle



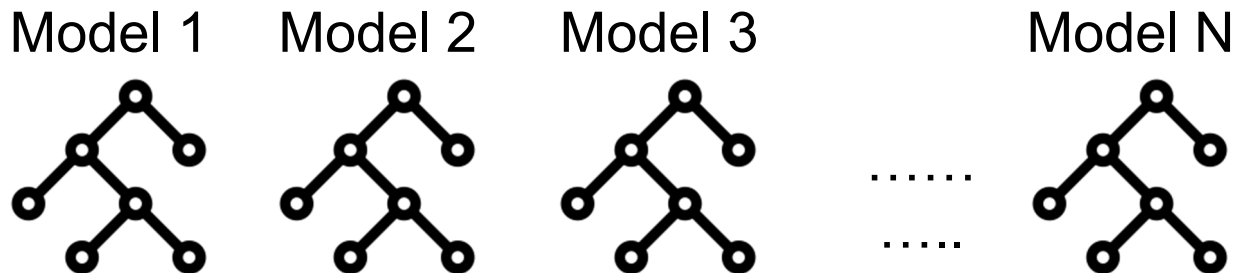
The Predictive Model Markup Language (PMML) is an XML-based predictive model interchange format



Model Versioning

One might generate many different models, how do we keep all kinds of models?

- Archive model with the same purpose under one file directory
- Persist model in a key/value store database



Archive Model in Folder Structure

/project

|-- models

| |-- model_1539753870

| | |-- metadata.json

| | |-- classifier.pkl

| |-- model_1539726970

| | |-- metadata.json

| | |-- classifier.pkl

| |-- model_1539777870

| | |-- metadata.json

| | |-- classifier.pkl

Unix Timestamp



Archive Model in a K/V Store

A key-value store, is a data storage paradigm designed for storing, retrieving, and managing associative arrays, a data structure more commonly known today as a dictionary or hash table.

Key	Value
1539753870	Model
1539726970	Model2
1539777870	Model3

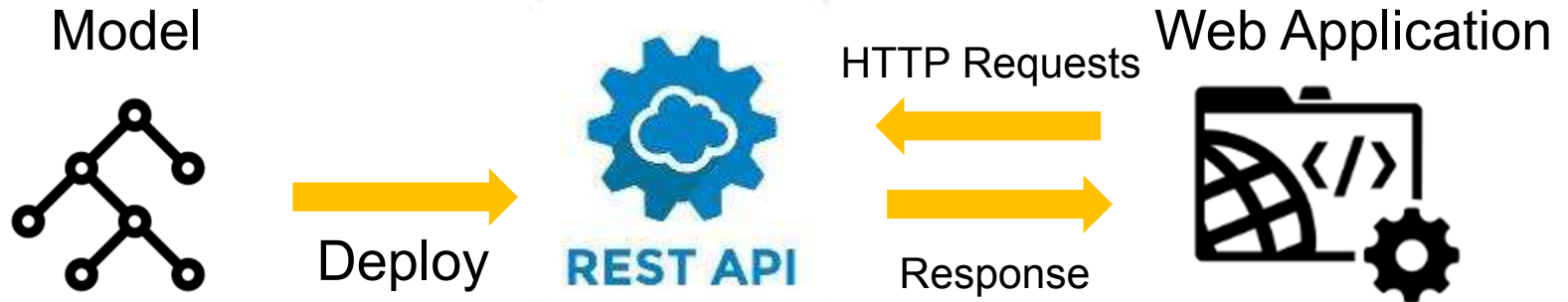
- Easily Accessible
- Consistency
- Central Management



Model Deployment

Serve Model With API

application programming interface (API) is a set of clearly defined methods of communication among various components



Build API With Various Language



python



Flask



Java™



Jersey

RESTful Web Services in Java.

Access Different Model Version

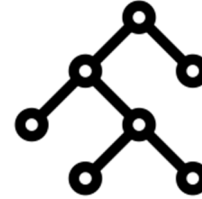
Sample API Code

```
from flask import Flask, jsonify, request
```

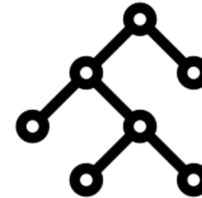
```
app = Flask(__name__)
```

```
@app.route('/predict',  
methods=["POST"])  
def predict(model_version):  
    model = load_model(model_version)  
    ...  
    pred = model.predict(data)  
    return jsonify(prediction)
```

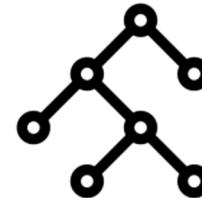
Model 3



Model 2

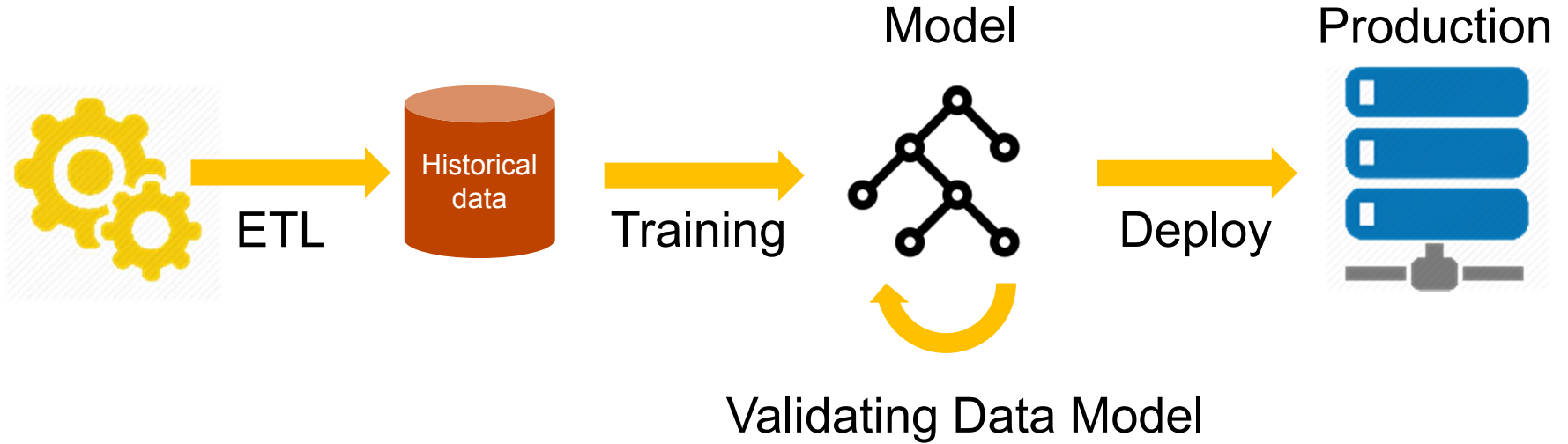


Model 1



Disaster
Recovery

Model Deployment





Docker

Docker performs operating-system-level virtualization

Lightweight virtual machines

Docker runs "containers", containers are isolated from each other and bundle their own tools, libraries and configuration files



DevOps

Continuous Integration

The practice of merging all developer working copies to a shared mainline several times a day

Continuous Delivery

Software engineering approach in which teams produce software in short cycles, ensuring that the software can be reliably released at any time

DevOps Tools



Jenkins



Travis CI



CODESHIP



GitLab

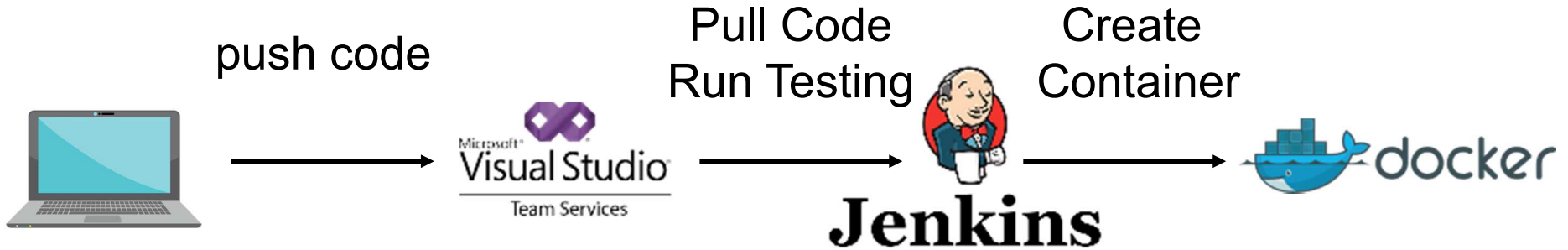


circleci



Bamboo

Automatic Deployment Process



Manage Multiple Dockers

Hyperparameter optimization

Model selection



Inference

Applying trained model to new data

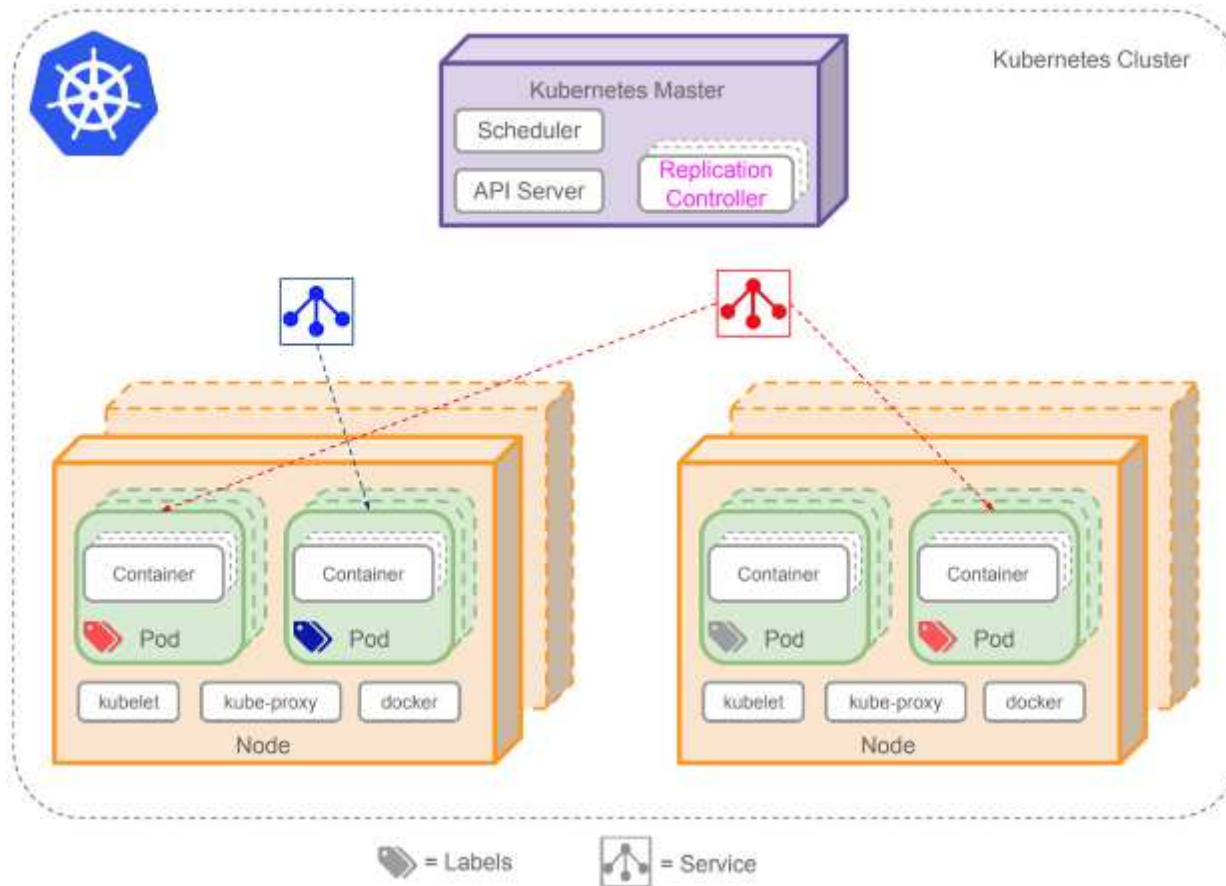


Scalability purpose

Scale the service



Kubernetes





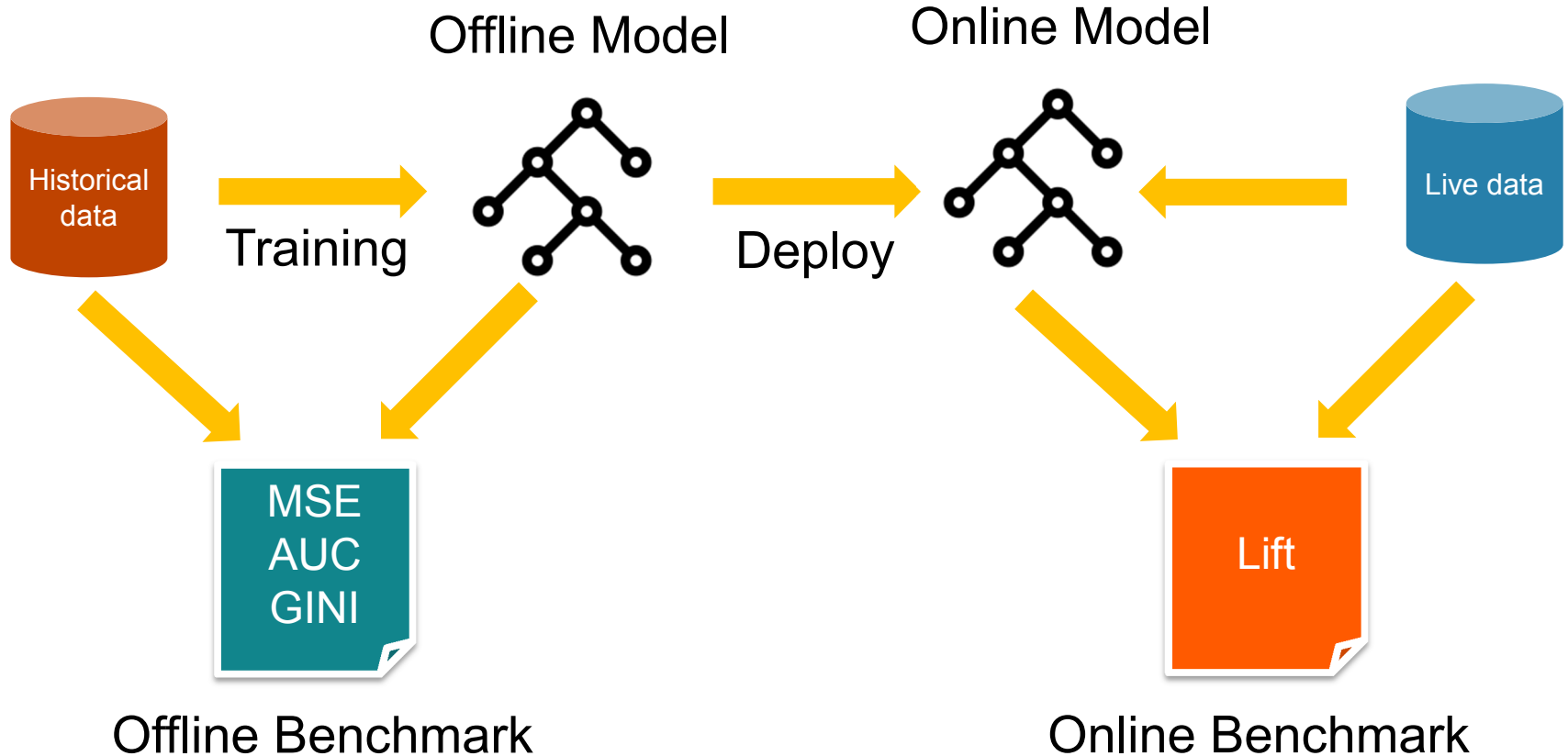
Evaluation



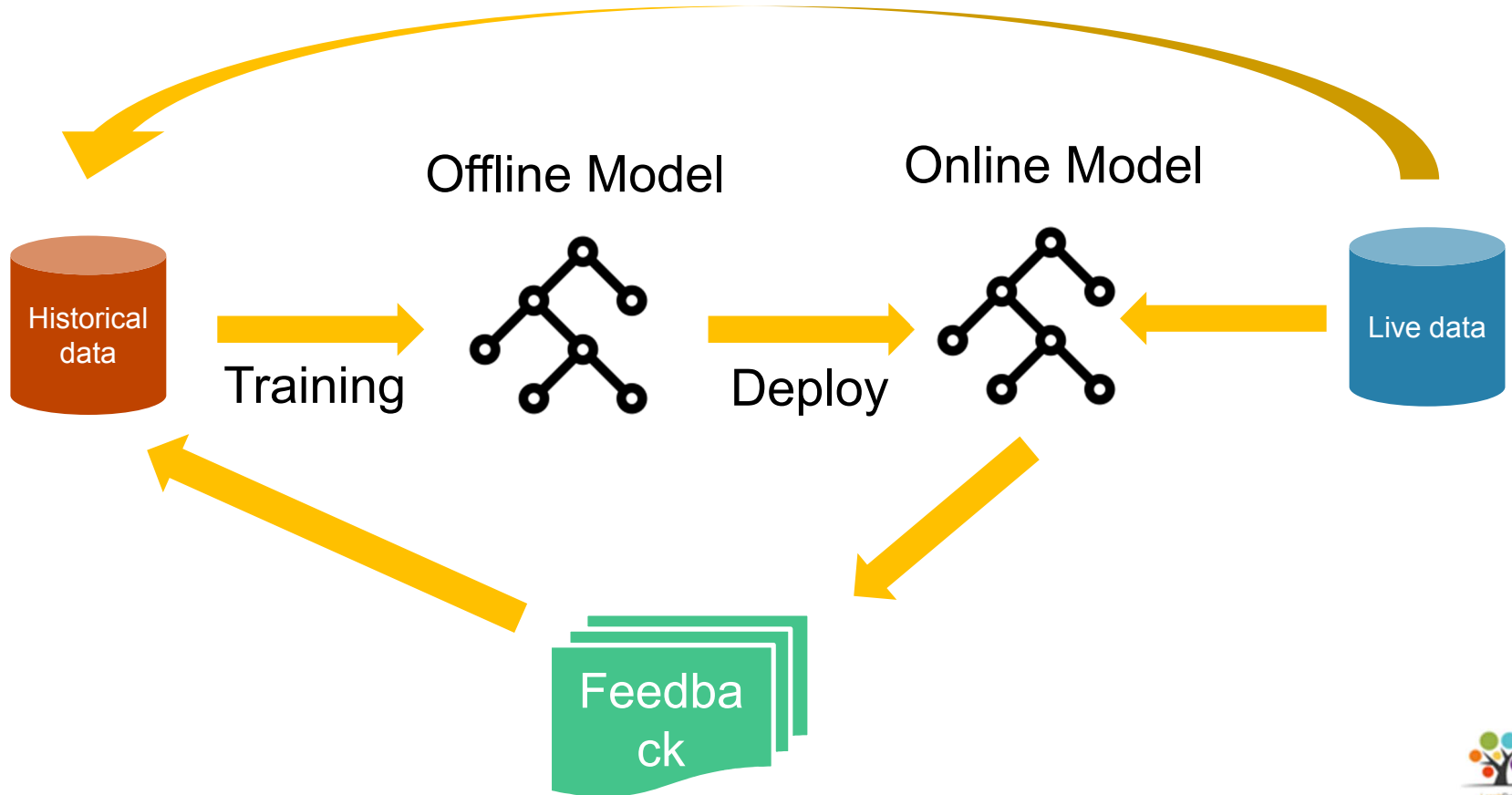
MSE, GINI, AUC

WHAT ARE YOU TALKING ABOUT?

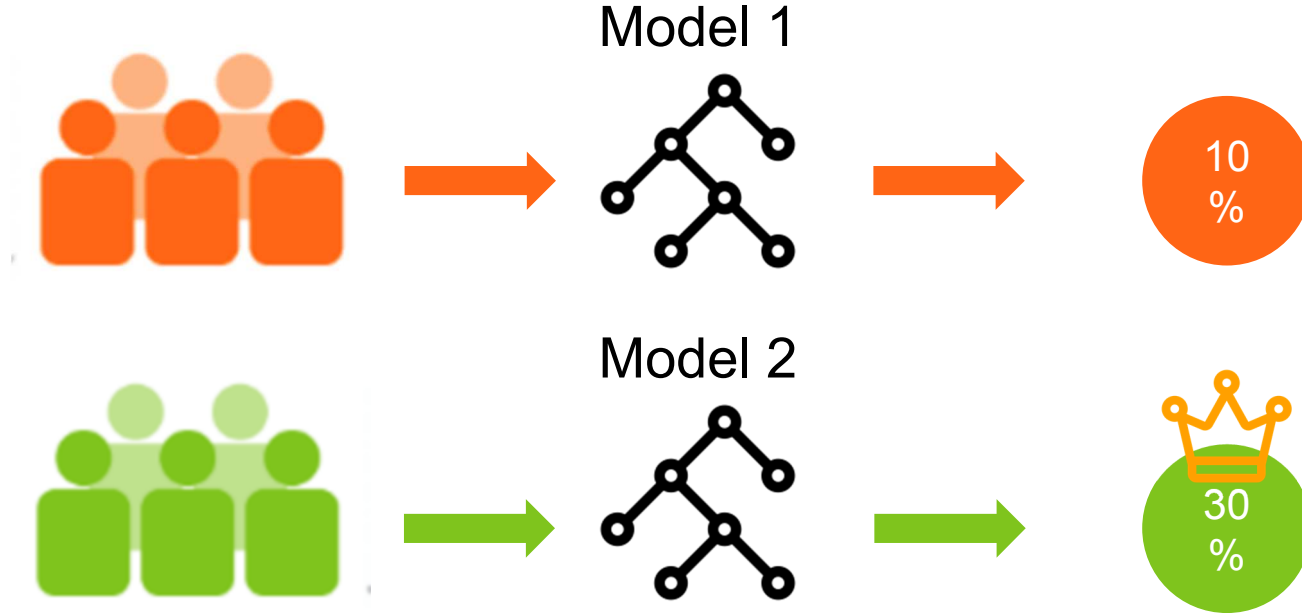
Online/Offline Benchmark



Collects Online Model Feedback



A/B Testing





Monitoring

Logging

Log Everything

CPU/DISK/Memory utilization

System Performance

- Response Latency
- Process Efficiency

Model Performance

- Online Evaluation
- Offline Evaluation

Deployment Process

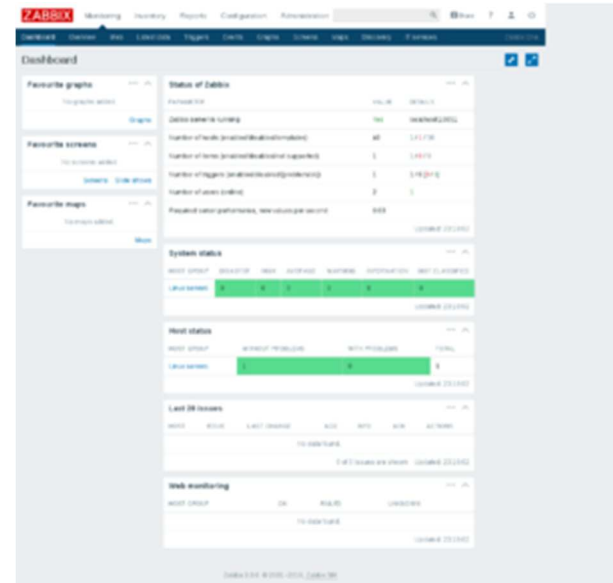
Zabbix

Open source monitoring software
tool for diverse IT components

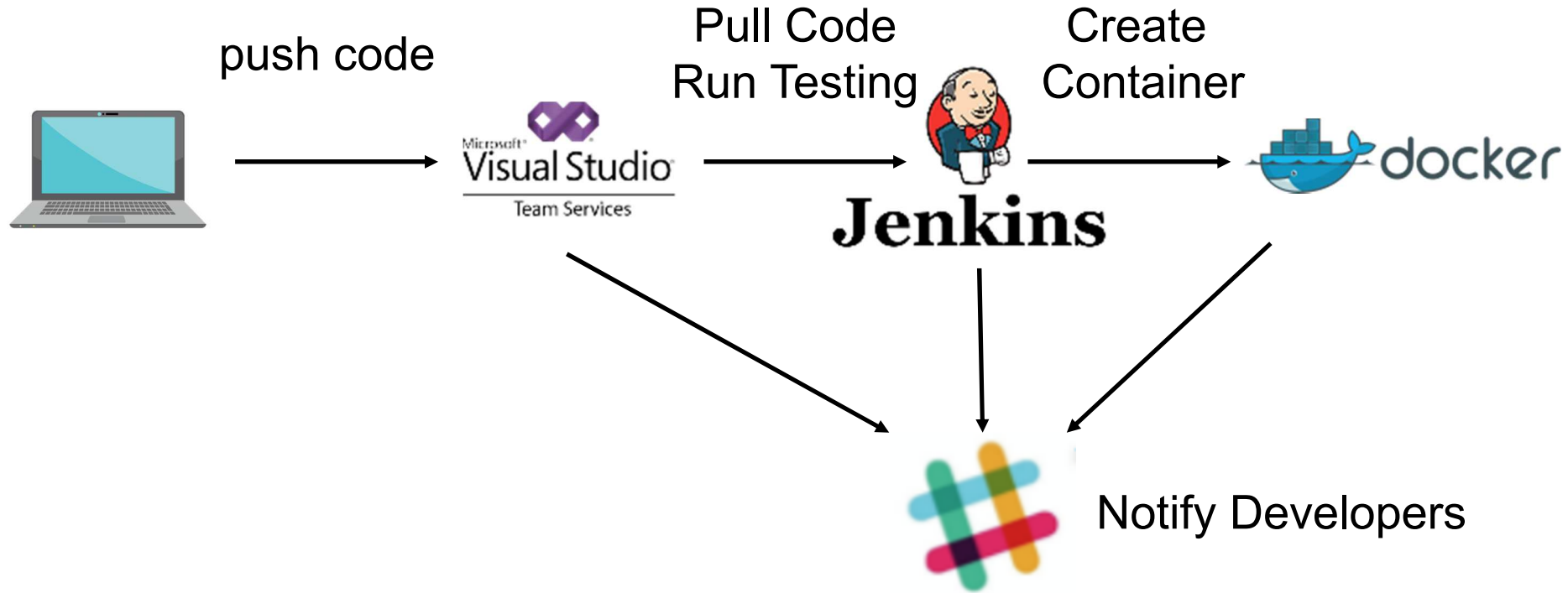
- Networks
- Servers
- Virtual machines
- Cloud services

Provides monitoring metrics

- Network utilization
- CPU load
- Disk space consumption



Slack





Enterprise Analysis Framework

Enterprise Analysis Framework

Relational Database

- Operation related system (Business Intelligence, Customer Relation Management System).
- Business Backbone

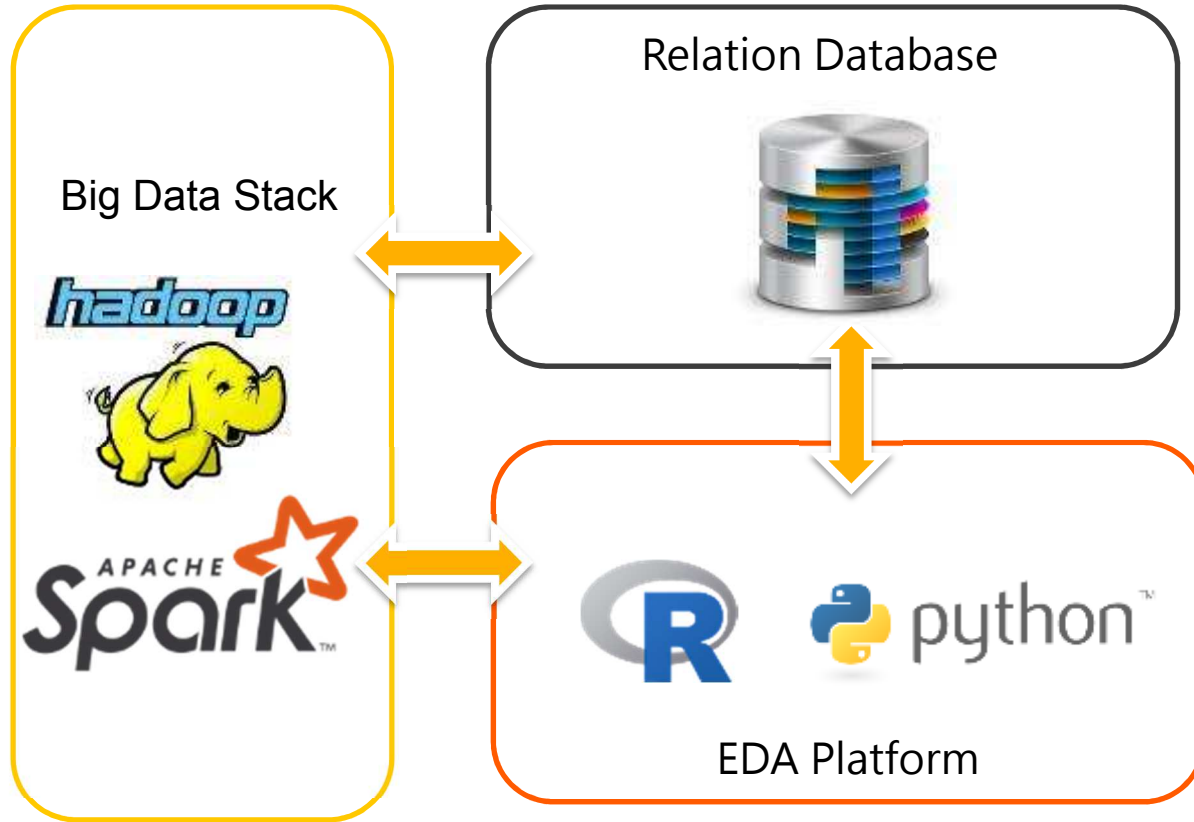
Exploratory Data Analysis

- Explore the possibility of data through exploration and analysis,
- Try to gain more insight into the data

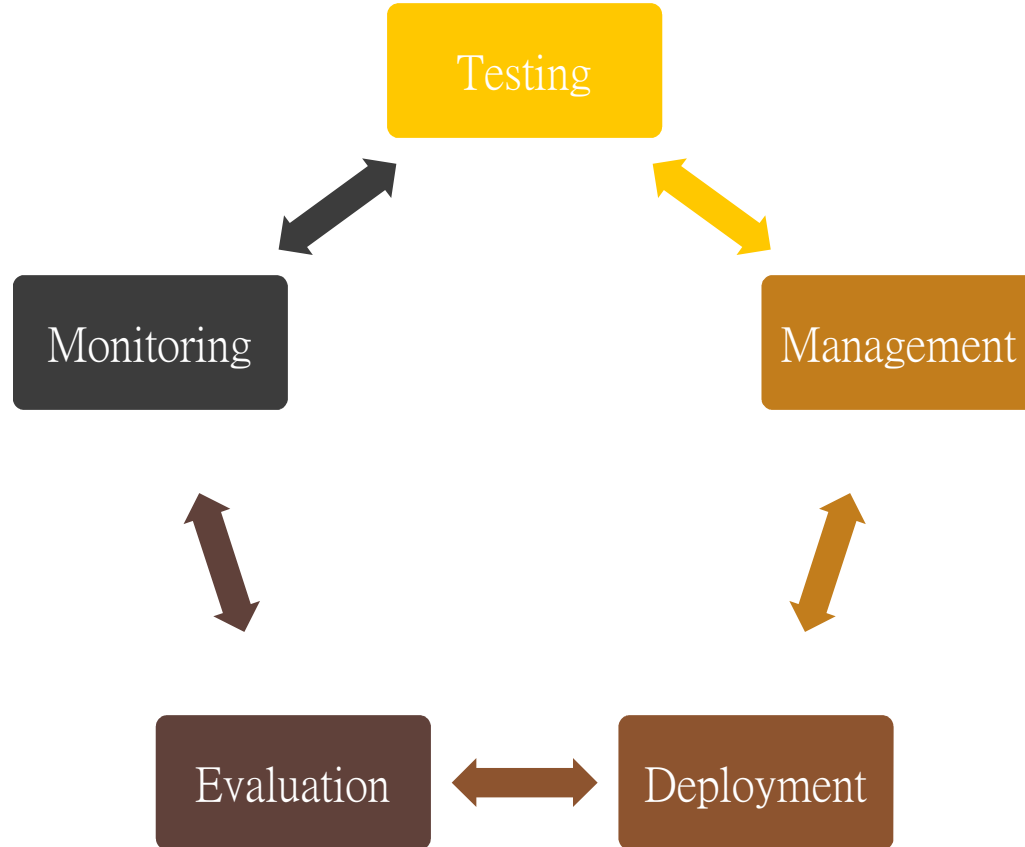
Big Data Stack

- Data preprocessing
- Provide low-cost data storage

Enterprise Analysis Framework



Go Production





THANK YOU

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Phone: 0929094381